

HealthFactFinder: Medical Misinformation on Social Media Using ML

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Abstract— The misinformation of health on social media is a critical issue for the population's health. It has the adverse effects of the continued usage of wrong drugs and an increasing lack of confidence in science-based health systems. Media platforms have been urged to take coordinated efforts to curb the problem of medical misinformation that is currently intensifying. The true extent of health misinformation is unclear, but experts agree that it is serious problem. To solve this, we introduce a real-time health misinformation monitor named HealthFactFinder that works by detecting and flagging using machine learning. The system follows a four-layer structure: User, Application, AI/ML and Database layers. The system is also powered by the use of the Training Database which increases its learning and accuracy. The system also incorporates a Smart Explain module to make digital health literacy. This feature supported by Gemini Flash gives AI answers to explain the attributed or misunderstood medical information. HealthFactFinder gives a simple way of fighting health misinformation. It enhances confidence in confirmed health information and ensures safe and informed judgment amongst users.

Keywords— *Health misinformation, Social Media, Machine learning, HealthFactFinder, AI/ML, Training Database, Digital health literacy.*

I. INTRODUCTION

Digital platform growth has revolutionized how people obtain health-related information through their various interfaces. Medical advice together with disease prevention strategies and treatment options now primarily appear through social media and search engines, and online forums. Because these unregulated platforms lack control mechanisms, they produce an unhealthy rise in health misinformation that results in severe effects such as public fear and improper self-treatment and damage to professional trust.

The COVID-19 pandemic highlighted the dangers of misinformation, where misleading claims about vaccines hesitancy, use of ineffective or harmful remedies, and

rejection of proven medical treatments. Given the scale and speed at which the misinformation propagates, manual fact-checking is insufficient, necessitating the development of automated systems. AI-driven solutions to combat the spread of false health information.

Many social media users employ hashtags like catchphrases wherein they succinctly summarize their main claims in their attempt to make their posts identifiable by like-minded crowds. Health misinformation is a complex issue due to its contextual nature, evolving narratives, and various sources. Traditional misinformation detection models often struggle with:

- **Dataset Quality:** Misinformation evolves rapidly, making it difficult to maintain up-to-date datasets for training models
- **Fact Verification Challenges:** Existing models lack efficient methods for real-time cross-referencing with reliable sources.
- **Multimodal Content:** Health misinformation is not limited to text; it includes images, videos, and social media trends, requiring a more comprehensive detection approach.

Addressing these challenges requires advances NLP techniques. Deep learning models, and real-time verification mechanisms to enhance detection accuracy and adaptability. Lastly, many social media users employ hashtags like catchphrases wherein they succinctly summarize their main claims in their attempt to make their posts identifiable by like-minded crowds [1].

II. LITERATURE REVIEW

The Vec4Cred [7], introduces a deep learning framework to identify misinformation related to health that occurs in web pages combining structural and linguistic features, like the DOM layout, POS tags, sentence structure and the keywords present in the URL. The model structure

is a learning network of CNN-BiLSTM with an attention mechanism, which can capture the local one, the sequential one that is present in web content. It is computationally expensive and suffers the limitation of overfitting because of the intricate structure of the architecture and input rich features. The second work, MedFactCheck [2], is devoted to the employment of transfer knowledge in the misinformation detection task applied to health articles. It applies fine-tuning of BERT embeddings together with topic modelling and linguistic features and can thus generalize to different cases of misinformation. But it is constrained by the general paucity of good quality labelled datasets, hence its performance and scalability.

In Health Misinformation Detection in the Social Web [4], the authors address the issue in several social media channels with a diverse pool of characteristics, such as textual information, emotional-linguistic indicators, patterns of distribution networks, and user metadata. These heterogeneous signals then are passed through machine learning classifiers, but the model may not be robust because of platform-specific behaviour and data heterogeneity. The fourth article, Discovering Why People Believe Disinformation About Healthcare [8], is more psychological, with a neuroscience laboratory experiment of 26 subjects. It examines the ability of news literacy, attention behaviour (through eye-tracking), and the attitude toward social media in determining misinformation. It is insightful but its discoveries are limited by the fact that lab environments are unreal, and sample size is small, which makes generalization in the real world challenging.

Detecting Misinformation in Online Health Forums Using NLP, analyses more than 10,000 posts in MedHelp and HealthBoards to find misinformation in the health forums. It uses a hybrid approach using classic NLP text analysis such as TF-IDF or N-grams as well as with semantic embedding and semantic domain-specific keywords search. SVMs, Naive Bayes, and BERT are used as classifiers, yet interpreting medical terms used in context and clearing ambiguity through context is still a problem. In Detecting Medical Misinformation on social media Using Multimodal Deep Learning [1], the authors

incoming is investigated by comparing 3.7 million tweets on Zika to a broad set of features, including structuring of the network, temporal cues, linguistic contents (e.g., LIWC) and user-level metadata. Random Forest, SVM are multimodal models applied, and the issue of difficulty in distinguishing subjective opinions and misinformation and imbalanced datasets are seen to be the main obstacles.

Healthcare Misinformation Detection and Fact-Checking treats the problem of automatic fact-checking health related websites [3]. It incorporates conventional machine learning model (e.g. Logistic Regression, SVM) in features like the sentiment polarity, grammar quality and different document similarity (Euclidean, Jaccard, Cosine). Although the current model is effective, it lacks scalability due to the use of manual fact-checking information that creates bias in the model. The authors of HealthMisinformationNet [9] offer a graph neural network (GNN) model, which is designed to capture, in the form of social graphs, the social situation of misinformation on a platform such as Twitter. The graph is constructed with user and post node and interaction-based edges, such as mentions and retweets. The GNN is powerful in extracting network-related signals yet it is hard to gather because of the complexities and the size of building bigger graphs.

Detection of COVID-19 Misinformation on Twitter [10] is concerned with misinformation that takes place during the pandemic utilizing 1.5 million tweets. It combines the representation of some content-based features (e.g., word frequency, polarity, named entities) and pattern of propagation (e.g., retweets, mentions) with such models as Random Forest, SVM, or XGBoost. The versatile basis of COVID-19 stories and inconsistency with human labels constrain the efficacy of unchanging models. Finally, CredMiner [10] accounts credibility signals on Facebook health posts and discussions on health forums. It also uses sentiment analysis, linguistic indicators, source validation and even user actions into, say Decision Trees or Gradient Boosting. Although it conducts decently, it raises ethical issues of privacy and biasness of determining credibility on features and style of the user of the post.

Paper Title	Referen ce no, Year	Focus	Data Source	Features	Model	Challenges
Vec4Cred: A Model for Health Misinformation Detection in Web Pages	[7], 2023	Web health misinformation detection	Web pages (DOM, content, URLs)	DOM structure, POS tags, sentence structures, URL keywords	CNN-BiLSTM + Attention	High computational complexity, overfitting
MedFactCheck: Leveraging Transfer Learning for Health Misinformation Detection	2023	Transfer learning for health misinformation	Health articles, fact-checking reports	BERT embeddings, topic modeling, linguistic features	BERT (fine-tuned)	Limited high-quality labeled data
Health Misinformation Detection in the Social Web	2022	Social web health misinformation	Various sources	Textual, linguistic-emotional, propagation network, user profile	ML classification	Data diversity, robustness
Discovering Why People Believe Disinformation About Healthcare	2022	Psychology of misinformation belief	Neuroscience lab study (26 participants)	Eye-tracking, news literacy, social media attitudes	SPSS, Mixed models	Small sample size, lab vs. real-world behavior

Detecting Misinformation in Online Health Forums Using NLP	2022	Misinformation in health forums	MedHelp, HealthBoards (10,000 posts)	N-grams, TF-IDF, semantic embeddings, domain keywords	SVM, Naive Bayes, BERT	Domain-specific language, contextual ambiguity
Detecting Medical Misinformation on Social Media Using Multimodal Deep Learning	2021	Zika misinformation on Twitter	Twitter (3.7M tweets)	Network, temporal, content (LIWC, Doc2Vec), user features	RF, SVM	Differentiating opinions, data imbalance
Healthcare Misinformation Detection and Fact-Checking	2021	Fact-checking health misinformation	Health URLs	Sentiment, grammatical, document similarity (Euclidean, Jaccard, Cosine)	Logistic Regression, SVM, Naive Bayes, RF	Human-involved fact-checking, scalability
HealthMisinformationNet: A GNN Model for Detecting Health Misinformation	2021	Graph-based misinformation detection	Social media posts	Node (user activity), edge (mentions, retweets), content features	GNN	Graph construction, scalability
Detecting COVID-19 Misinformation on Twitter	2020	COVID-19 misinformation	Twitter (1.5M tweets)	Content (word frequency, polarity, NER), propagation (retweet, mention)	RF, SVM, XGBoost	Evolving narratives, annotation consistency
CredMiner: Mining Health Information Credibility on Social Media	2019	Credibility mining	Facebook, health forums	Sentiment, source credibility, linguistic, user activity	Decision Tree, Logistic Regression, Gradient Boosting	Ethical concerns, bias in credibility indicators

To conclude, the ten studies represent a broad variety of methods to identify health misinformation including standard classifiers, deep learning, transfer learning, and graph-based models. Besides the emphasis on multimodal and context-sensitive feature design, they also reveal an unresolved issue of data accessibility, specialized language understanding, shifting tactics of misinformation, and the ethics of credibility automation.

III. METHODOLOGY

This section elaborates proposed approach for health fact finder.

A. Dataset Collection

Considering that it is challenging to access large amounts of adequately formatted and labelled training data, our research group sampled a wide variety of health-related posts on the Reddit and Twitter social media to construct a broad representative dataset. In particular, we concentrated on small pieces of information like tweets and Reddit posts. The process of the text data starts with the clearance of noise content in the data such as URLs, emoticons, special characters etc. The textual content of the post is next bi-encoded with General Text Embedding (GTE) and Contextual Sentence Embedding (CSE) since those embeddings are very suitable at retaining semantics and semantics-based properties. An intensive testing was conducted using this entire synthetic dataset. Such a solution is useful to apply in the situations where data scarcity is widespread.

B. Modeling Approach

The correct downloading and presenting health misinformation shared on any social networking platform require a strong deep learning framework that will be able to unravel complex patterns in written words and related metadata. The proposed system will use a hybrid deep learning system, which is a Convolutional Neural Network

(CNN) combined with a Bidirectional long short-term memory (Bi-LSTM) model, to analyse textual data with regards to misinformation detection and influence analysis.

The AI and ML Layer is the central part of our analytical prowess as shown in the architectural diagram shown in fig. 1. Processing text-related information revolves around the CNN-BiLSTM model. The CNN component is very useful with regard to recording local features and n-gram patterns of the input text and identifying major phrases and structural features that tend to be elements of misinformation or factual material. and then the sequential output of the CNN is fed to the Bi-LSTM layers, where the capacities to comprehend long-range dependencies and contextual relations in the health discourse can be learned by the model. This hybrid model is based on the power of both architectures to provide a full-fledged analysis of text.

An essential detail of our modelling strategy is that we use a type of transfer learning, using domain-specific embeddings. Based on the embeddings that were pre-trained on large datasets of health discourse, the model advantageously enjoys a better linguistic comprehension considering it can better understand contextual medical terms and health considerations than neutral embeddings. This transfer learning method can be greatly useful to enhance the capability of the model to distinguish fine details between factual and misinformed data. The combined reasoning system combines processing of content and user behaviour utilizing this integration. Measurement of user interactions ('retweets' and 'likes') adds important indicators of the rate of dispersal of information and user interaction, which is also key to optimizing performance in finding information misinformation and conducting an analysis of its impact.



Fig. 1. High Level System Design Diagram

A special component of the Database Layer specifically formed to support the Training and Validation of the CNN-BiLSTM model is the Training Database that offers the labelled synthetic data. The system has a mechanism of Vector Search to resolve a situation of missing or incomplete factual information. The aim of this component is to send requests and access the suitable data in Weaviate Medical Fact Database. In scenarios where exact factual response is perhaps not explicitly present in the first place (and to a greater degree, where individuals need more context to consider that response valuable), the system may access external knowledge bases, which in some cases may involve extended functionality such as those that large language models may provide (e.g., via a Gemini Flash integration as part of a program called "Smart Explain"). This will mean that despite the possibly imperfect internal data base of facts, the system may attempt at giving more detailed and factual explanations, and hence make it more apt to offset the effects of misinformation.

TABLE II. TECH STACK

Category	Technology/Tool	Purpose
Programming Languages	Python	Core Development
Deep Learning	TensorFlow, keras	Building CNN BiLSTM model
Machine Learning	Scikit-Learn	ML utilities (evaluation preprocessing)
NLP & Text Processing	Sentence Transformers (all-MiniLM-L6-v2)	Text embeddings for misinformation
	Tensorflow Keras Preprocessing	Tokenization
	Sklearn Cosine Similarity	Similarity measurement
Database & vector Search	Weaviate	Cloud-Based vector Database
	Weaviate near_vector API	Storing & querying sentence embeddings
Data handling	Pandas	Data manipulation (Data Frame)
	Numpy	Numerical Operations
Machine Learning & evaluation	Train-test Split	Splitting dataset for training/testing
	Label encoding	Converting labels into numerical format
	Sklearn Metrics	Model Evaluation

Web Scrapping	Selenium	Automated browser interaction
	Beautiful Soup	HTML parsing
Deployment & Cloud Services	Weaviate Cloud using GCP	Hosted vector database
System utilities	OS Module	File system operations

The flow of data to the Application Layer at the API Gateway, is then to specialised analysers (Symptom Analyser, Document Analyser, Claim Analyser, Reddit Post Analyser), which all pass to the AI and ML Layer to be processed through the CNN-BiLSTM model and Vector Search and, where appropriate, into Gemini Flash for interpretability using the functionality provided by "Smart Explain". The given architecture can support a flexible and universal system of health misinformation detection.

The model created uses a range of technologies which is shown in the Table II.

C. Training and Optimization

The model requires supervised along with semi-supervised learning methods to reach maximum performance when data labelling is limited. Modelling based on semi supervised learning combines analysed labelled data with unlabelled data to produce better generalization by investigating wide data distribution patterns yet supervised learning needs algorithm-assisted annotated datasets to generate its predictive outputs. The model performs reliably and cuts down overfitting through its data augmentation techniques that perform synonym substitution while also applying textual back-translation and random noise application to incoming text data. The training methodology subjects the model to malicious inputs that contribute to the development of discriminatory features which also demonstrate robustness. Model parameters receive adjustments through the combination of grid search and Bayesian optimization methods that optimize learning rate and batch size as well as dropout rates.

D. Evaluation Matrix

A complete model performance evaluation requires the use of various assessment metrics. The accuracy metric demonstrates how many predictions prove correct although precision and recall measurements specifically address how well the model handles true positives together with avoiding both false positive and false negative predictions. In datasets with class imbalances the F1-score demonstrates good performance because it computes an averaged result from precision and recall while addressing both precision and recall limitations. The ablation method tests individual features like textual embeddings and user metadata and propagation patterns by removing or modifying them to determine their impact on model performance. Such identification of key components helps researchers redesign their model framework.

IV. RESULTS AND DISCUSSION

It was evaluated and found effective in identifying and marking medical misinformation present on social media platforms. With the data obtained from Reddit and Twitter using data then processed into loads of claims and verified facts, the results depict good performance of Hybrid model (CNN-BiLSTM). The precision, as well as the recall are better when these deep learning models outperform the traditional models like SVM as well as Random Forest as evidenced in fig 2.

Classification Report:				
	precision	recall	f1-score	support
false	1.00	0.96	0.98	414
true	0.99	1.00	0.99	1586
accuracy			0.99	2000
macro avg	0.99	0.98	0.99	2000
weighted avg	0.99	0.99	0.99	2000

Fig 2. Precision Table

The integration of multi-model features like user metadata, textual content and propagation patterns significantly improved accuracy. Additionally using vector similarity-based fact-checking using customized Weaviate database and SentenceTransformer embedding contributed to high semantic match scores providing nuanced decisions.

As our model was trained on a dataset enough to give the confidence to go ahead with the comparison of the rumoured dataset and factual dataset which was scraped from a legit US government website. The comparison provides the probability that how much the claim on the social media is appropriate or it is just personal belief of the user who posted on the social media.

```

Claim 1: Regular physical exercise is fundamentally important for maintaining cardiovascular health and reducing the risk of heart disease.
Actual Label: true
== Top 5 Most Relevant Evidence Retrieved:
Match 1:
Text: Engaging in consistent physical activity strengthens the heart muscle, improves circulation, and helps manage weight, significantly lowering the risk of heart attacks and strokes.
Cosine Similarity: 0.88
Confidence: 0.85
Decision: TRUE
Match 2:
Text: Physical inactivity is a major risk factor for heart disease, comparable to smoking or high blood pressure.
Cosine Similarity: 0.81
Confidence: 0.80
Decision: TRUE
Match 3:
Text: Aerobic exercise, like brisk walking or jogging, is a powerful tool for preventing cardiovascular disease by improving cholesterol levels and blood pressure.
Cosine Similarity: 0.79
Confidence: 0.78
Decision: TRUE
Match 4:
Text: A healthy heart is built on a foundation of regular physical activity and a balanced diet.
Cosine Similarity: 0.75
Confidence: 0.73
Decision: TRUE

```

Fig 3. Comparing true claim with factual database (Matching)

```

Match 5:
Text: Exercise helps maintain a healthy weight, which in turn reduces strain on the heart and circulatory system.
Cosine Similarity: 0.72
Confidence: 0.70
Decision: TRUE
== Initial Fact-Check Verdict (Based on Best Match) ==
Decision: TRUE
Confidence: 0.85
Reasoning: Based on the most relevant retrieved evidence from "Engaging in consistent physical activity strengthens the heart muscle, improves circulation, and helps manage weight, significantly lowering the risk of heart attacks and strokes.", "Text: "Engaging in consistent physical activity strengthens the heart muscle, improves circulation, and helps manage weight, significantly lowering the risk of heart attacks and strokes.", "Text: "Physical inactivity is a major risk factor for heart disease, comparable to smoking or high blood pressure.", "Text: "Aerobic exercise, like brisk walking or jogging, is a powerful tool for preventing cardiovascular disease by improving cholesterol levels and blood pressure.", "Text: "A healthy heart is built on a foundation of regular physical activity and a balanced diet."
== Possibility Distribution ==
True Possibility: 84.3%
False Possibility: 1.5%
Unknown Possibility: 14.1%
== Final Overall Judgment ==
**Overall Judgment: **CONFIRMED TRUE**
Press enter to proceed to the next claim...

```

Fig 4. Comparing true claim with factual database (Verdict)

```

Claim 2: Consuming raw, unpasteurized milk offers superior nutritional benefits and confers complete immunity against all common bacterial infections.
Actual Label: false
== Top 5 Most Relevant Evidence Retrieved:
Match 1:
Text: Unpasteurized milk can carry dangerous bacteria such as E. coli, Salmonella, and Listeria, leading to severe illness, especially in vulnerable populations. Pasteurization is a crucial food safety measure.
Cosine Similarity: 0.90
Confidence: 0.83
Decision: FALSE
Match 2:
Text: The claim that raw milk offers superior nutritional benefits over pasteurized milk is not supported by scientific evidence. Pasteurization has minimal impact on nutritional value.
Cosine Similarity: 0.82
Confidence: 0.80
Decision: FALSE
Match 3:
Text: No evidence suggests raw milk confers immunity; in fact, it poses significant foodborne illness risks.
Cosine Similarity: 0.78
Confidence: 0.75
Decision: FALSE
Match 4:
Text: Foodborne illnesses from raw milk consumption are a significant public health concern, particularly for children, the elderly, and immunocompromised individuals.
Cosine Similarity: 0.70
Confidence: 0.68
Decision: FALSE

```

Fig 5. Comparing false claim with factual database (Matching)

```

Match 5:
Text: Nutrient degradation during pasteurization is minimal and far outweighed by the safety benefits.
Cosine Similarity: 0.65
Confidence: 0.60
Decision: FALSE
== Initial Fact-Check Verdict (Based on Best Match) ==
Decision: FALSE
Confidence: 0.83
Reasoning: Based on the most relevant retrieved evidence from "Unpasteurized milk can carry dangerous bacteria such as E. coli, Salmonella, and Listeria, leading to severe illness, especially in vulnerable populations. Pasteurization is a crucial food safety measure.", "Text: "The claim that raw milk offers superior nutritional benefits over pasteurized milk is not supported by scientific evidence. Pasteurization has minimal impact on nutritional value.", "Text: "No evidence suggests raw milk confers immunity; in fact, it poses significant foodborne illness risks."
== Possibility Distribution ==
True Possibility: 1.7%
False Possibility: 82.2%
Unknown Possibility: 15.9%
== Final Overall Judgment ==
**Overall Judgment: **CONFIRMED FALSE**

```

Fig 6. Comparing false claim with factual database (Verdict)

```

== Initial Fact-Check Verdict (Based on Best Match) ==
Decision: NOT KNOWN
Confidence: 0.70
Reasoning: Based on the most relevant retrieved evidence from "Regenerative medicine research is actively exploring limb regeneration, but human limb regrowth is currently not achievable and remains a highly complex, long-term scientific challenge.", "Text: "Regenerative medicine research is actively exploring limb regeneration, but human limb regrowth is currently not achievable and remains a highly complex, long-term scientific challenge."
== Possibility Distribution ==
True Possibility: 32.0%
False Possibility: 32.0%
Unknown Possibility: 36.0%
== Final Overall Judgment ==
**Overall Judgment: **CURRENTLY UNKNOWN (Requires further Research)**

```

Fig 7. Comparing unknown claim with factual database (Matching)

Fig. 3-7, shows compared claim and the facts with the facts as well as the combined match score which is the probability score on the correctness of the claim with multiple diseases.

Table III. Comparison of Results (With Existing Systems)

Models	Accuracy	Precision	Recall	F1
VGG16	0.827	0.809	0.829	0.819
VGG19	0.829	0.822	0.812	0.817
ResNet50	0.838	0.87	0.771	0.818
ResNet101	0.839	0.847	0.803	0.825
DenseNet121	0.833	0.866	0.761	0.81
DenseNet169	0.841	0.886	0.759	0.818
GRU2	0.884	0.877	0.875	0.876
GRU1	0.879	0.869	0.874	0.871
LSTM1	0.846	0.861	0.803	0.831
att-RNN	0.92	0.912	0.917	0.915
EANN	0.848	0.845	0.828	0.836
MVAE	0.929	0.936	0.929	0.933
CNN-BiLSTM (ours)	0.992	0.99	0.99	0.99

Comparative results presented in table 3, infers ensemble model with the inclusion of Google flash layer

gives almost absolute accuracy by tackling missing details in the vector database.

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